SLIP-20 B: Write a C program to display all file from current directory and its subdirectory whose size is greater than 'n' Bytes

where n is accepted from user through command line

```
#include"stdio.h"

#include"stdlib.h"

#include"unistd.h"

#include"fcntl.h"

#include"sys/stat.h"

#include"sys/types.h"

#include"dirent.h"

int main(int argc,char*argv[])

{

int n;

DIR *d;

struct dirent *de;

struct stat s;

printf("\n Enter value of n(size)=>");

scanf("%d",&n);

d=opendir(".");

if(d==NULL)

{

printf("unable to open dir");

exit(0);

}

while((de=readdir(d))!=NULL)

{

stat(de->d_name,&s);

if(s.st_size>n)

{

printf("%s\t%d\n",de->d_name,s.st_size);

}

}

return 0;
```

```
}
//OUTPUT
/*
[fy@localhost Rohit]$ gedit acc13.c
[fy@localhost Rohit]$ gcc acc13.c;
[fy@localhost Rohit]$ ./a.out
 Enter value of n(size)=>10000
[fy@localhost Rohit]$ ./a.out

 Enter value of n(size)=>100
.           269
..          4096
practical1.c        837
bhole     126
practical2.c        660
practical3.c        1055
prac4.c   2028
ass13.c   333
acc13.c   471
a.out     8680

*/
```

SLIP-01 A: Write a C program to find whether a given file is present in cuurent directory or not

```
#include "stdio.h"
#include "stdlib.h"
#include "unistd.h"
#include "fcntl.h"
#include "sys/types.h"
#include "sys/stat.h"
#include "dirent.h"
int main()
{
char fname[30];
int fd;

printf("Enter file name=");
scanf("%s",fname);
fd=open(fname,O_RDONLY);
if(fd==-1)
{
printf("File is not found");
}
else
{
printf("File is present in the current directory");
return 0;
}
}
/*
Output:
[fy@localhost Rohit AOS]$ gedit sleep1.c
[fy@localhost Rohit AOS]$ gcc sleep1.c
[fy@localhost Rohit AOS]$ ls
a.out   bhole  d.txt     p4.txt       practical2.c   sleep1.c
```

```
a.txt   b.txt  p4ab.txt  prac4.c       practical3.c

bb.txt  c.txt  p4a.txt   practical1.c  practicl2.txt

[fy@localhost Rohit AOS]$ ./a.out bb

Enter file name=bb.txt

File is present in the current directory[fy@localhost Rohit AOS]$

[fy@localhost Rohit AOS]$ ./a.out b

Enter file name=b.txt

File is present in the current directory[fy@localhost Rohit AOS]$ ./a.out

Enter file name=ajhfsjA.C

File is not found[fy@localhost Rohit AOS]$ ./a.out

Enter file name=PRAC4.C

File is not found[fy@localhost Rohit AOS]$

[fy@localhost Rohit AOS]$

[fy@localhost Rohit AOS]$

[fy@localhost Rohit AOS]$ ./a.out

Enter file name=prac4.c

File is present in the current directory[


*/
```

SLIP-02 A:Wrute c program that a string as an argument and return akk file that begins with that name in the current directory

```
#include "stdio.h"
#include "stdlib.h"
#include "unistd.h"
#include "fcntl.h"
#include "sys/types.h"
#include "sys/stat.h"
#include "dirent.h"
int main(int argc,char *argv[])
{
int c=0;
DIR *d;
d=opendir(".");
struct dirent *dr;
if(d==NULL)
{
        printf("unable nto open directory");
    exit(0);
    }
   while((dr=readdir(d))!=NULL)
        {
                printf("%s\t",dr->d_name);
                c++;
                }
                printf("\n No of files=>%d",c);
                closedir(d);
                return 0;
}
```

/*[fy@localhost Rohit]$ gedit practical3.c

[fy@localhost Rohit]$ gcc practical3.c

```
[fy@localhost Rohit]$ ls -l

total 36

-rwxrwxr-x. 1 fy fy 8576 Mar 22 15:56 a.out

-rw-rw-r--. 1 fy fy   28 Mar 22 15:53 bb.txt

-rwxrwxr-x. 1 fy fy  126 Mar 22 14:49 bhole

-rw-rw-r--. 1 fy fy  837 Mar 22 15:23 practical1.c

-rw-rw-r--. 1 fy fy  660 Mar 22 15:55 practical2.c

-rw-rw-r--. 1 fy fy  451 Mar 22 15:56 practical3.c

-rw-rw-r--. 1 fy fy   23 Mar 22 15:14 practicl2.txt

[fy@localhost Rohit]$ ls -l |wc -l

8

[fy@localhost Rohit]$ ./a.out

.	..	practical1.c	bhole	practical2.c	practicl2.txt	practical3.c	bb.txt	a.out


*/
```

SLIP-03 A:Write a C program to find fike properties such as inode number ,number of hard link, file permission, file size

and modification time so on of a given file using stat() system call

```
#include"stdio.h"

#include"unistd.h"

#include"fcntl.h"

#include"sys/stat.h"

#include"sys/types.h"

#include"time.h"

int main()

{

char fn[100];

char *s;

int fd;

struct stat s1;

printf("Enter file name=");

scanf("%s",fn);

stat(fn,&s1);

printf("File name=%s\n",fn);

printf("Inod number=%d\n",s1.st_ino);

printf("number of hard link= %d\n",s1.st_nlink);

printf("File size =%d\n",s1.st_size);

printf("file Permision=");

if(S_IRUSR & s1.st_mode)

printf("Read");

else

printf("-------");

if(S_IWUSR & s1.st_mode)

printf("Write");

else

printf("-------");

if(S_IXUSR & s1.st_mode)

printf("Exicute");
```

```
else
printf("-------");
//group of file
if(S_IRGRP & s1.st_mode)
printf("Read");
else
printf("-------");
if(S_IWGRP & s1.st_mode)
printf("Write");
else
printf("-------");
if(S_IXGRP & s1.st_mode)
printf("Exicute");
else
printf("-------");
if(S_IROTH & s1.st_mode)
printf("Read");
else
printf("-------");
if(S_IWOTH & s1.st_mode)
printf("Write");
else
printf("-------");
if(S_IXOTH & s1.st_mode)
printf("Exicute");
else
printf("-------");
printf("\n");
s=ctime(&s1.st_atime);
printf("File access time =%s\n",s);
s=ctime(&s1.st_mtime);
printf("File modification time =%s\n",s);
return 0
```

SLIP-04 A:Write a C program to find like properties such as inode number ,number of hard link, file permission, file size and modification time so on of a given file using fstat() system call

```
#include"stdio.h"
#include"unistd.h"
#include"fcntl.h"
#include"sys/stat.h"
#include"sys/types.h"
#include"time.h"
int main()
{
char fn[100];
char *s;
int fd;
struct stat s1;
printf("Enter file name=");
scanf("%s",fn);
fd=open(fn,O_RDONLY);
fstat(fd,&s1);
printf("File name=%s\n",fn);
printf("Inod number=%d\n",s1.st_ino);
printf("number of hard link= %d\n",s1.st_nlink);
printf("File size =%d\n",s1.st_size);
printf("file Permision=");
if(S_IRUSR & s1.st_mode)
printf("Read");
else
printf("-------");

if(S_IWUSR & s1.st_mode)
printf("Write");
else
printf("-------");
if(S_IXUSR & s1.st_mode)
```

```
printf("Exicute");
else
printf("-------");
//group of file
if(S_IRGRP & s1.st_mode)
printf("Read");
else
printf("-------");
if(S_IWGRP & s1.st_mode)
printf("Write");
else
printf("-------");
if(S_IXGRP & s1.st_mode)
printf("Exicute");
else
printf("-------");
if(S_IROTH & s1.st_mode)
printf("Read");
else
printf("-------");
if(S_IWOTH & s1.st_mode)
printf("Write");
else
printf("-------");
if(S_IXOTH & s1.st_mode)
printf("Exicute");
else
printf("-------");
printf("\n");
s=ctime(&s1.st_atime);
printf("File access time =%s\n",s);

s=ctime(&s1.st_mtime);
printf("File modification time =%s\n",s);
return 0;
```

```
}
/*
[fy@localhost Rohit AOS]$ gedit sleep2.c
[fy@localhost Rohit AOS]$ gcc sleep2.c
[fy@localhost Rohit AOS]$ ls
a.out   bhole  d.txt     p4.txt       practical2.c   sleep1.c
a.txt   b.txt  p4ab.txt  prac4.c      practical3.c   sleep2.c
bb.txt  c.txt  p4a.txt   practical1.c  practicl2.txt
[fy@localhost Rohit AOS]$ ./a.out
Enter file name=a.txt
File name=a.txt
Inod number=12128
number of hard link= 1
File size =3
file Permision=ReadWrite-------ReadWrite-------Read--------------
File access time =Tue Mar 22 16:21:14 2022

File modification time =Tue Mar 22 16:16:11 2022

[fy@localhost Rohit AOS]$ stat a.txt
  File: 'a.txt'
  Size: 3                Blocks: 8          IO Block: 4096   regular file
Device: fd02h/64770d     Inode: 12128       Links: 1
Access: (0664/-rw-rw-r--)  Uid: ( 1000/     fy)   Gid: ( 1000/     fy)
Context: unconfined_u:object_r:user_home_t:s0
Access: 2022-03-22 16:21:14.083568000 +0530
Modify: 2022-03-22 16:16:11.773497000 +0530
Change: 2022-03-22 16:21:14.083568565 +0530
 Birth: -

*/
```

SLIP-05 A:Write a C program to create an unnamed pipe The child proccess will write following three messsages to pipe and parent

process display it

Message1="Hello World"

Message2="Hello SPPU"

Message3="Linux is funny"

```
#include "stdio.h"

#include "stdlib.h"

#include "dirent.h"

#include "fcntl.h"

int main (int argc,char *argv[])

{

int i;

int p[2],n;

char buf[70];

char *m1="1.Hellow";

char *m2="2.Good buy";

char *m3="3.Exit";

if(pipe(p)<0)

exit(0);

n=fork();

if(n<0)

{

 printf("Fork failed");

exit(0);

}

if(n==0)//chiled process

{

  write(p[1],m1,70);

  write(p[1],m2,70);

  write(p[1],m3,70);

}

else

{

 for(i=0;i<3;i++)
```

```
{
 read(p[0],buf,70);
 printf("%s \n",buf);
}
}
return 0;
}
/*

Output=

[fy@localhost ~]$ gedit assnew1.c
[fy@localhost ~]$ gcc assnew1.c
[fy@localhost ~]$ ./a.out
1.Hellow
2.Good buy
3.Exit

*/
```

SLIP-05 B: Write a C program to read all txt files in the current directory and merge them all to one txt file and return a file descriptor for the new file

```
#include"stdio.h"
#include"stdlib.h"
#include"dirent.h"
#include"fcntl.h"
#include"string.h"
#include"sys/types.h"
int main()
{
int fd,fd1,l;
DIR *d;
char ch;
char new[100];
struct dirent *de;
printf("\n enter the file");
scanf("%s",new);
d=opendir(".");
if(d==NULL)
{
printf("open dir unable");
exit(0);
}
fd1=open(new,O_RDWR|O_CREAT,0666);
if(fd1<0)
{
printf("unable open new file");
exit(0);
}
while((de=readdir(d))!=NULL)
{
l=strlen(de->d_name);
if(de->d_name[l-4]=='.'&& de->d_name[l-3]=='t'&& de->d_name[l-2]=='x'&& de->d_name[l-1]=='t')
{
fd=open(de->d_name,O_RDONLY);
```

```
while(read(fd,&ch,1))
write(fd1,&ch,1);

}
close(fd);
}
}
/*fy@localhost abhi]$ gcc t1.c
 * [fy@localhost abhi]$ ./a.out
 *
 *  enter the file bgs
 *  [fy@localhost abhi]$ cat b1.txt
 *   bhki
 *   adjnjjhfj
 *   56459+45
 *   [fy@localhost abhi]$ cat d.txt
 *     cya
 *      abhi
 *       fuxk
 *       [fy@localhost abhi]$ cat bgs

 *       [fy@localhost abhi]$ ls
 *       abhi   armstrong.py  b1.txt  d      endtxt.c  fact.py  lcm.py  p2.c    perfact.py  pipe.c     prime.py
 *       a.out  b1           bgs     d.txt  env.c     gcd.py   p1.c    pal.py  pipe1.c     prime1.py  t1.c
 *       [fy@localhost abhi]$ cat bgs
 *
 *
 *       [fy@localhost abhi]$ gedit t1.c
 *       [fy@localhost abhi]$ gcc t1.c
 *       [fy@localhost abhi]$ ./a.out
 *
 *        enter the file bgs
 *        [fy@localhost abhi]$ cat b1.txt
 *         bhki
 *         adjnjjhfj
```

```
*       56459+45
*       [fy@localhost abhi]$ cat d.txt
*         cya
*          abhi
*           fuxk
*           [fy@localhost abhi]$ cat bgs
*             cya
*              abhi
*               fuxk
*                bhki
*                adjnjjhfj
*                56459+45
*/
```

SLIP-18 B:Write a C program to implement the following unix\linux command . your program should block the signal ctrl-c amd ctrl-\signal during the execution ls-l|wc-l

```c
#include "stdio.h"
#include "stdlib.h"
#include "fcntl.h"
#include "unistd.h"
#include "signal.h"

void sh(int sig)
{
printf("\n signal cought");
fflush(stdout);
}
int main()
{
int p[2],n;
signal(SIGINT,sh);
signal(SIGQUIT,sh);
pipe(p);
n=fork();
if(n==0)
{
sleep(5);
close(p[0]);
close(1);
dup(p[1]);
close(p[1]);
execlp("ls","ls","-l",NULL);
}
else
{
wait();
close(p[1]);
close(0);
dup(p[0]);
close(p[0]);
execlp("wc","wc","-l",NULL);
}
```

SLIP NO -10 A: Write a c program to diaplay statistics related to memory allocation system.(use mallocinfo syatem call)

```
#include "stdio.h"
#include "stdlib.h"
#include "malloc.h"
int main()
{
char *c;
struct mallinfo m;
m=mallinfo();
printf("\n Non_mapped space allocated=>%d",m.arena);
printf("\n number of free chunks=>%d",m.ordblks);
printf("\n Number of free fastbin bloks=>%d",m.smblks);
printf("\n Number of mmapped regions=>%d",m.hblks);
printf("\n space allocated in mmapped regions=>%d",m.hblkhd);
printf("\n See below=>%d",m.usmblks);
printf("\n Space in freed fastbin bloacks=>%d",m.fsmblks);
printf("\n Total allocated space=>%d",m.fordblks);
printf("\n Total free space=>%d",m.fordblks);
printf("\n Top-most,relesable space=>%d",m.keepcost);
return 0;
}
```

OUTPUT:

```
[fy@localhost ~]$ gedit malinfo.c
[fy@localhost ~]$ gcc malinfo.c
[fy@localhost ~]$ ./a.out

 Non_mapped space allocated=>0
 number of free chunks=>1
 Number of free fastbin bloks=>0
 Number of mmapped regions=>0
 space allocated in mmapped regions=>0
 See below=>0
 Space in freed fastbin bloacks=>0
 Total allocated space=>0
 Total free space=>0
 Top-most,relesable space=>0[fy@localhost ~]$
```

SLIP NO-17 A:Write a C program to display the given msg 'n' times (make a use of setjmp AND LONGJMP system-repeat call)

```
#include "stdio.h"
#include "stdlib.h"
#include "setjmp.h"
#include "string.h"
int main()
{
char m[50];
int n,c=0;
jmp_buf buf;
printf("\n enter message to display=>");
gets(m);
printf("\n enter n no of times=>");
scanf("%d",&n);
setjmp(buf);
printf("%s\n",m);
c++;
if(c<n)
longjmp(buf,1);
return 0;
}
```

OUTPUT:

```
[fy@localhost ~]$ gedit message.c
[fy@localhost ~]$ gcc message.c
[fy@localhost ~]$ ./a.out

 enter message to display=>hv desai

 enter n no of times=>5
hv desai
hv desai
hv desai
hv desai
hv desai
[fy@localhost ~]$
```

SLIP-06 A:Write a C program to map a given file in memory and the contain of mapped file in reverse

```
#include "stdio.h"
#include "stdlib.h"
#include "fcntl.h"
#include "sys/stat.h"
#include "sys/types.h"
#include "sys/mman.h"

int main(int argc,char* argv[])
{
int fd,size,i;
struct stat s1;
char *f,ch;
if(argc<2)
{
printf("\n Enter file name on command line");
exit(0);
}
fd=open(argv[1],O_RDONLY);
if(fd<0)
{
printf("\n unable to open file");
exit(0);
}
stat(argv[1],&s1);
size=s1.st_size;
f=(char*)mmap(0,size,PROT_READ,MAP_PRIVATE,fd,0);
for(i=size-1;i>=0;i--)
{
ch=f[i];
printf("%c",ch);
}
return 0;
}
```

```
o/p:[fy@localhost ~]$ gcc rev.c
[fy@localhost ~]$ ./a.out
[fy@localhost ~]$ ./a.out v11.txt

evahb lahsiv[fy@localhost ~]$ gedit rev.c
```

SLIP NO-12 A: Write a c program to send SIGALRM signal by child process to parent process and parent process make a provision to catch the signal and display alarm is fired.(use kill,fork,signal and sleep system call)

```
#include "stdio.h"
#include "stdlib.h"
#include "fcntl.h"
#include "signal.h"

void sh(int sig)
{
printf("\n Signal cought");
fflush(stdout);
}
int main()
{
int n;
signal(SIGINT,sh);
n=fork();
if(n==0)
{
printf("\n in child process");
sleep(20);
kill(getppid(),SIGALRM);
printf("\n child existing");
sleep(5);
}
else
{
wait();
printf("\n parent exiting");
}
return 0;
}
```

```
output:
[fy@localhost ~]$ gedit s12.c
[fy@localhost ~]$ gcc s12.c
[fy@localhost ~]$ ./a.out

^C in child process
 Signal cought

Alarm clock
```

SLIP NO -15 B:Write a c program which creates child process and child process catches a sighup sigint singquit the parent send a sighup or sighint after every 3 sec at end of 15 sec parent send sighquit signal to child and child terminates by displaying msg my papa has killed me

```
#include "stdio.h"
#include "stdlib.h"
#include "fcntl.h"
#include "signal.h"

void sh1()
{
signal(SIGHUP,sh1);
printf("\n SIGHUP cought");
}

void sh2()
{
signal(SIGINT,sh2);
printf("\n SIGINT,sh2");
}

void sh3()
{
printf("\n My papa has killed me");
exit(0);
}

int main()
{
int n;
n=fork();
signal(SIGHUP,sh1);
signal(SIGINT,sh2);
signal(SIGQUIT,sh3);
if(n<0)
{
printf("unable to create child");
exit(0);
}
if(n==0)
{
signal(SIGHUP,sh1);
signal(SIGINT,sh2);
signal(SIGQUIT,sh3);
for(;;)
;
}
```

```
else
{
printf("\nin parent");
kill(n,SIGINT);
sleep(3);
kill(n,SIGHUP);
sleep(3);
kill(n,SIGINT);
sleep(3);
kill(n,SIGHUP);
sleep(3);
kill(n,SIGHUP);
sleep(3);
kill(n,SIGQUIT);
sleep(3);
wait();
}
return 0;
}

OUTPUT:

[fy@localhost ~]$ gedit s15b.c
[fy@localhost ~]$ gcc s15b.c
[fy@localhost ~]$ ./a.out


 SIGINT,sh2
 SIGHUP cought
 SIGINT,sh2
 SIGHUP cought
 SIGHUP cought
 My papa has killed mein parent[fy@localhost ~]$ gedit s15b.c
```

SLIP NO-20 A: Write a c program that print the exit status of a terminated child process

```
#include "stdio.h"
#include "stdlib.h"
#include "fcntl.h"
#include "signal.h"

int main()
{
int n,status;
n=fork();
if(n==0)
{
execlp("date","date","-l",NULL);
}
else
{
waitpid(n,&status,0);
if(WIFEXITED(status));
{
int e;
e=WEXITSTATUS(status);
printf("exit status =>%d",e);
}
}
return 0;
}
```

OUTPUT:

```
exit status =>1[fy@localhost ~]$ gedit s20
[fy@localhost ~]$ gcc s20.c
s20.c: In function ‘main’:
s20.c:12:1: warning: incompatible implicit declaration of built-in function
‘execlp’ [enabled by default]
 execlp("date","date","-l",NULL);
 ^
[fy@localhost ~]$ ./a.out
date: invalid option -- 'l'
Try 'date --help' for more information.
exit status =>1[fy@localhost ~]$
```

SLIP NO-16 A:Write a c program that catches the ctrl-c (SIGINT) signal for the first time and display the approriate message and exit on pressing ctrl-c again

```
#include "stdio.h"
#include "stdlib.h"
#include "fcntl.h"
#include "signal.h"

void sh(int sig)
{
printf("\n SIGINT signal cought");
signal(SIGINT,SIG_DFL);
fflush(stdout);
}
int main()
{
signal(SIGINT,sh);
while(1)
{
}
return 0;
}
```

OUTPUT:

```
[fy@localhost ~]$ gedit signal.c
[fy@localhost ~]$ gcc signal.c
[fy@localhost ~]$ ./a.out
^C
 SIGINT signal cought
```

SLIP NO-06 B: Write aC program that behave like a shell it has its own prompt say NewShell$. any normal shell command is executed from your shell by styarting a child process .....

```
#include"stdio.h"
#include"stdlib.h"
#include"fcntl.h"
#include"unistd.h"
#include"sys/types.h"
#include"sys/stat.h"
#include"dirent.h"
void list(char ch,char *dname);
int main()
{
        char comm[80],t1[20],t2[20],t3[20],t4[20];
        int n;
        system("Clear");
        while(1)
        {
                printf("Newshell$");
                fflush(stdin);
                gets(comm);
                n=sscanf(comm,"%s%s%s%s",t1,t2,t3,t4);
                switch(n)
                {
        case 1:
                        if(!fork())
                         {
                           execlp(t1,t1,NULL);
                           perror(t1);
                         }
                         break;
        case 2:
                        if(!fork())
                         {
                          execlp(t1,t1,t2,NULL);
                           perror(t1);
```

```
 }
                                    break;

            case 3:
                                  if(!fork())
                                    {
                    if(strcmp(t1,"list")==0)
                    list(t2[0],t3);
                    else
                    {

                                      execlp(t1,t1,t2,t3,NULL);
                                      perror(t1);
                                      }
                                    }
                                    break;

                        case 4:

                                  if(!fork())
                                    {
                                      execlp(t1,t1,t2,t3,t4,NULL);
                                      perror(t1);
                                    }
                                    break;
                 }
               }
               }
void list(char ch,char *dname)
{
  struct stat s;
  struct dirent *de;
  DIR *d;
  int c=0;
  d=opendir(dname);
```

```
 if(d==NULL)
 {
  printf("Unable to open directory");
  exit(0);
  }
  switch(ch)
  {
   case 'f':
          while((de=readdir(d))!=NULL)
                 printf("%s\n",de->d_name);
          break;
   case 'n':
          while((de=readdir(d))!=NULL)
          c=c+1;
          printf("\n no of Entries =>%d",c);
          break;
   case 'i':
          while((de=readdir(d))!=NULL)
          {
               struct stat s;
               char c[100];
               strcpy(c,dname);
               strcat(c,"/");
               strcat(c,de->d_name);
               stat(c,&s);
               printf("\n%s\t%d",de->d_name,s.st_ino);
          }
           break;


          }
          }
```

Output=

```
[fy@localhost RJ(AOS)]$ ./a.out
sh: Clear: command not found
Newshell$list i KGF
```

```
Newshell$
.           271377030
..          403052099
kgf1.c      271377032Newshell$
Newshell$list f KGF
Newshell$.
..
kgf1.c
Newshell$list n KGF
Newshell$
 no of Entries =>3Newshell$
Newshell$
```

SLIP N0 -08 B:Write a C program which receive fike name as command line argument and displaythose filename in ascending order according

tp their sizes

```
#include "stdio.h"

#include "stdlib.h"

#include "unistd.h"

#include "fcntl.h"

#include "sys/types.h"

#include "sys/stat.h"

#include "string.h"

int main(int argc,char *argv[])

{

int i,j;

struct stat s1,s2;

char *s;

if(argc<2)

{

printf("insuficient argument");

exit(0);

}

for(i=1;i<argc;i++)

for(j=i+1;j<argc;j++)

{

stat(argv[i],&s1);

stat(argv[j],&s2);

if(s1.st_size > s2.st_size)

{

strcpy(s,argv[i]);

strcpy(argv[i],argv[j]);

strcpy(argv[j],s);

}

}

for(i=1;i<argc;i++)

{

stat(argv[i],&s1);
```

```
printf("%s\t%d\n",argv[i],s1.st_size);
}
return 0;
}
/*[fy@localhost Rohit]$ gedit prac4.c
[fy@localhost Rohit]$ gcc prac4.c
[fy@localhost Rohit]$ gedit p4.txt
[fy@localhost Rohit]$ gedit p4a.txt
[fy@localhost Rohit]$ gedit p4ab.txt
[fy@localhost Rohit]$ ls -l p*.*
-rw-rw-r--. 1 fy fy   14 Mar 22 16:13 p4ab.txt
-rw-rw-r--. 1 fy fy   14 Mar 22 16:13 p4a.txt
-rw-rw-r--. 1 fy fy   35 Mar 22 16:13 p4.txt
-rw-rw-r--. 1 fy fy  545 Mar 22 16:12 prac4.c
-rw-rw-r--. 1 fy fy  837 Mar 22 15:23 practical1.c
-rw-rw-r--. 1 fy fy  660 Mar 22 15:55 practical2.c
-rw-rw-r--. 1 fy fy 1055 Mar 22 15:59 practical3.c
-rw-rw-r--. 1 fy fy   23 Mar 22 15:14 practicl2.txt
[fy@localhost Rohit]$ gcc prac4.c
[fy@localhost Rohit]$ ./a.out p4.txt p4a.txt p4ab.txt
p4ab.txp4a.txtpt     35
p4a.txtpt 35
t             35
[fy@localhost Rohit]$
[fy@localhost Rohit]$
[fy@localhost Rohit]$
[fy@localhost Rohit]$ gedit a.txt
[fy@localhost Rohit]$ gedit b.txt
[fy@localhost Rohit]$ gedit c.txt
[fy@localhost Rohit]$ gedit d.txt
[fy@localhost Rohit]$ ls -l p*.*
-rw-rw-r--. 1 fy fy   14 Mar 22 16:13 p4ab.txt
-rw-rw-r--. 1 fy fy   14 Mar 22 16:13 p4a.txt
-rw-rw-r--. 1 fy fy   35 Mar 22 16:13 p4.txt
-rw-rw-r--. 1 fy fy  545 Mar 22 16:12 prac4.c
```

```
-rw-rw-r--. 1 fy fy  837 Mar 22 15:23 practical1.c

-rw-rw-r--. 1 fy fy  660 Mar 22 15:55 practical2.c

-rw-rw-r--. 1 fy fy 1055 Mar 22 15:59 practical3.c

-rw-rw-r--. 1 fy fy   23 Mar 22 15:14 practicl2.txt

[fy@localhost Rohit]$ gcc prac4.c

[fy@localhost Rohit]$ ./a.out a.txt b.txt c.txt d.txt

d.txt        2

a.txt        3

b.txt        8

c.txt        23


*/
```

SLIP NO-09 A:Write a C program to display as well as resets the environment varible such as path,home,root etc

```
#include "stdio.h"
#include "stdlib.h"
#include "dirent.h"
#include "fcntl.h"

int main (int argc,char *argv[],char *env[])
{
int i;
printf("Envirmental Variable  with values => \n");
for(i=0;env[i]!=NULL;i++)
        printf("\n %s",env[i]);
        printf("\n Reseting value of home,Root,Path =>\n");
        setenv("HOME","BGS",1);
        setenv("ROOT","xyz",1);
        setenv("PATH","lmn",1);
printf("\n Envirmental variable after reset =>\n");
printf("\n Root=> %s",getenv("ROOT"));
printf("\n Path=> %s",getenv("PATH"));
printf("\n Home=> %s",getenv("HOME"));
return 0;
}
/*
[fy@localhost ~]$ gcc ass15.c
[fy@localhost ~]$ ./a.out
Envirmental Variable  with values =>
 XDG_VTNR=1
 SSH_AGENT_PID=2905
 XDG_SESSION_ID=1
 HOSTNAME=localhost.localdomain
 IMSETTINGS_INTEGRATE_DESKTOP=yes
 VTE_VERSION=5204
 TERM=xterm-256color
```

SHELL=/bin/bash

XDG_MENU_PREFIX=gnome-

HISTSIZE=1000

PERL5LIB=/home/fy/perl5/lib/perl5:

GNOME_TERMINAL_SCREEN=/org/gnome/Terminal/screen/c503df66_8433_438b_b0b5_6fdfb1e318ff

QTDIR=/usr/lib64/qt-3.3

QTINC=/usr/lib64/qt-3.3/include

PERL_MB_OPT=--install_base /home/fy/perl5

IMSETTINGS_MODULE=none

QT_GRAPHICSSYSTEM_CHECKED=1

USER=fy

LS_COLORS=rs=0:di=38;5;27:ln=38;5;51:mh=44;38;5;15:pi=40;38;5;11:so=38;5;13:do=38;5;5:bd=48;5;232;38;5;11:cd=48;5;232;38;5;3:or=48;5;232;38;5;9:mi=05;48;5;232;38;5;15:su=48;5;196;38;5;15:sg=48;5;11;38;5;16:ca=48;5;196;38;5;226:tw=48;5;10;38;5;16:ow=48;5;10;38;5;21:st=48;5;21;38;5;15:ex=38;5;34:*.tar=38;5;9:*.tgz=38;5;9:*.arc=38;5;9:*.arj=38;5;9:*.taz=38;5;9:*.lha=38;5;9:*.lz4=38;5;9:*.lzh=38;5;9:*.lzma=38;5;9:*.tlz=38;5;9:*.txz=38;5;9:*.tzo=38;5;9:*.t7z=38;5;9:*.zip=38;5;9:*.z=38;5;9:*.Z=38;5;9:*.dz=38;5;9:*.gz=38;5;9:*.lrz=38;5;9:*.lz=38;5;9:*.lzo=38;5;9:*.xz=38;5;9:*.bz2=38;5;9:*.bz=38;5;9:*.tbz=38;5;9:*.tbz2=38;5;9:*.tz=38;5;9:*.deb=38;5;9:*.rpm=38;5;9:*.jar=38;5;9:*.war=38;5;9:*.ear=38;5;9:*.sar=38;5;9:*.rar=38;5;9:*.alz=38;5;9:*.ace=38;5;9:*.zoo=38;5;9:*.cpio=38;5;9:*.7z=38;5;9:*.rz=38;5;9:*.cab=38;5;9:*.jpg=38;5;13:*.jpeg=38;5;13:*.gif=38;5;13:*.bmp=38;5;13:*.pbm=38;5;13:*.pgm=38;5;13:*.ppm=38;5;13:*.tga=38;5;13:*.xbm=38;5;13:*.xpm=38;5;13:*.tif=38;5;13:*.tiff=38;5;13:*.png=38;5;13:*.svg=38;5;13:*.svgz=38;5;13:*.mng=38;5;13:*.pcx=38;5;13:*.mov=38;5;13:*.mpg=38;5;13:*.mpeg=38;5;13:*.m2v=38;5;13:*.mkv=38;5;13:*.webm=38;5;13:*.ogm=38;5;13:*.mp4=38;5;13:*.m4v=38;5;13:*.mp4v=38;5;13:*.vob=38;5;13:*.qt=38;5;13:*.nuv=38;5;13:*.wmv=38;5;13:*.asf=38;5;13:*.rm=38;5;13:*.rmvb=38;5;13:*.flc=38;5;13:*.avi=38;5;13:*.fli=38;5;13:*.flv=38;5;13:*.gl=38;5;13:*.dl=38;5;13:*.xcf=38;5;13:*.xwd=38;5;13:*.yuv=38;5;13:*.cgm=38;5;13:*.emf=38;5;13:*.axv=38;5;13:*.anx=38;5;13:*.ogv=38;5;13:*.ogx=38;5;13:*.aac=38;5;45:*.au=38;5;45:*.flac=38;5;45:*.mid=38;5;45:*.midi=38;5;45:*.mka=38;5;45:*.mp3=38;5;45:*.mpc=38;5;45:*.ogg=38;5;45:*.ra=38;5;45:*.wav=38;5;45:*.axa=38;5;45:*.oga=38;5;45:*.spx=38;5;45:*.xspf=38;5;45:

GNOME_TERMINAL_SERVICE=:1.108

SSH_AUTH_SOCK=/run/user/1000/keyring/ssh

SESSION_MANAGER=local/unix:@/tmp/.ICE-unix/2775,unix/unix:/tmp/.ICE-unix/2775

USERNAME=fy

GNOME_SHELL_SESSION_MODE=classic

PATH=/usr/lib64/qt-3.3/bin:/home/fy/perl5/bin:/usr/local/bin:/usr/local/sbin:/usr/bin:/usr/sbin:/bin:/sbin:/home/fy/.local/bin:/home/fy/bin

MAIL=/var/spool/mail/fy

DESKTOP_SESSION=gnome-classic

QT_IM_MODULE=ibus

XDG_SESSION_TYPE=x11

PWD=/home/fy

XMODIFIERS=@im=ibus

LANG=en_IN.UTF-8

GDM_LANG=en_IN.UTF-8

KDEDIRS=/usr

GDMSESSION=gnome-classic

HISTCONTROL=ignoredups

XDG_SEAT=seat0

HOME=/home/fy

SHLVL=2

GNOME_DESKTOP_SESSION_ID=this-is-deprecated

PERL_LOCAL_LIB_ROOT=:/home/fy/perl5

XDG_SESSION_DESKTOP=gnome-classic

LOGNAME=fy

QTLIB=/usr/lib64/qt-3.3/lib

XDG_DATA_DIRS=/home/fy/.local/share/flatpak/exports/share/:/var/lib/flatpak/exports/share/:/usr/local/share/:/usr/share/

DBUS_SESSION_BUS_ADDRESS=unix:abstract=/tmp/dbus-9QQhJ70Drk,guid=576888787f34ba675124914562553bb6

LESSOPEN=||/usr/bin/lesspipe.sh %s

WINDOWPATH=1

XDG_RUNTIME_DIR=/run/user/1000

DISPLAY=:0

QT_PLUGIN_PATH=/usr/lib64/kde4/plugins:/usr/lib/kde4/plugins

XDG_CURRENT_DESKTOP=GNOME-Classic:GNOME

PERL_MM_OPT=INSTALL_BASE=/home/fy/perl5

COLORTERM=truecolor

XAUTHORITY=/run/gdm/auth-for-fy-rG5Sk4/database

_=./a.out

Reseting value of home,Root,Path =>

Envirmental variable after reset =>

Root=> xyz

Path=> lmn

Home=> BGS[

*/

SLIPNO- 11 A:Write a C program to create variable length arrays using alloc () system call

```
#include"stdio.h"
#include"stdlib.h"
#include"alloca.h"
int main()
{
  int **a;
  int i,j,c[5],r;
  int count=0;
  printf("Enter number of rowes = ");
  scanf("%d",&r);
  a=(int**)alloca(sizeof(int)*r);
  for(i=0;i<r;i++)
  {
  printf("\n Enter number columns of row  %d  =  ",i);
  scanf("%d",&c[i]);
  }
  for(i=0;i<r;i++)
  {
    a[i]=(int*)alloca(sizeof(int)*c[i]);
  }
  printf("Enter element in array=\n");
  for(i=0;i<r;i++)
 {
  printf("\n Enetr %d row=>",i);
 for(j=0;j<c[i];j++)
  {
    printf("Enter %d column=>",j);
    scanf("%d",&a[i][j]);
  }
}
printf("Element of array is =>\n");
  for(i=0;i<r;i++)
 {
printf("\n");
```

```
 for(j=0;j<c[i];j++)
  {
     printf("%d\t",a[i][j]);
  }
}
}
```

Output

```
[fy@localhost RJ(AOS)]$ gcc allock.c
[fy@localhost RJ(AOS)]$ ./a.out
Enter number of rowes = 3
 Enter number columns of row  0  =  2
 Enter number columns of row  1  =  3
 Enter number columns of row  2  =  4
Enter element in array=
 Enetr 0 row=>Enter 0 column=>1
Enter 1 column=>10
 Enetr 1 row=>Enter 0 column=>2
Enter 1 column=>20
Enter 2 column=>30
 Enetr 2 row=>Enter 0 column=>4
Enter 1 column=>10
Enter 2 column=>20
Enter 3 column=>30
Element of array is =>
1       10
2       20      30
4       10      20      30
```

SLIP-12 B: Write a C program to display all file from current directory and its subdirectory whose size is greater than 'n' Bytes

where n is accepted from user through command line

```
#include"stdio.h"

#include"stdlib.h"

#include"unistd.h"

#include"fcntl.h"

#include"sys/stat.h"

#include"sys/types.h"

#include"dirent.h"

int main(int argc,char*argv[])

{

int n;

DIR *d;

struct dirent *de;

struct stat s;

printf("\n Enter value of n(size)=>");

scanf("%d",&n);

d=opendir(".");

if(d==NULL)

{

printf("unable to open dir");

exit(0);

}

while((de=readdir(d))!=NULL)

{

stat(de->d_name,&s);

if(s.st_size>n)

{

printf("%s\t%d\n",de->d_name,s.st_size);

}

}

return 0;

}
```

```
//OUTPUT
/*
[fy@localhost Rohit]$ gedit acc13.c
[fy@localhost Rohit]$ gcc acc13.c;
[fy@localhost Rohit]$ ./a.out

 Enter value of n(size)=>10000
[fy@localhost Rohit]$ ./a.out

 Enter value of n(size)=>100
.           269
..          4096
practical1.c        837
bhole       126
practical2.c        660
practical3.c        1055
prac4.c     2028
ass13.c     333
acc13.c     471
a.out       8680
```

SLIP NO-13 A:Write a C program that redirect standard output.txt(use dup and open system)

```
#include"stdio.h"
#include"stdlib.h"
#include"fcntl.h"
#include"dirent.h"
#include"unistd.h"
int main()
{
int fds,fdi;
fdi=open("d.txt",O_RDWR|O_CREAT,0666);
if(fdi<0)
{
printf("unable open file");
exit(0);
}
fds=dup(1);
if(dup2(fdi,1)<0)
{
printf("enable to duplicate");
exit(0);
}
printf("\n cya ");
printf("\n abhi");
printf("\n fuxk");
close(fdi);
fflush(stdout);
dup2(fds,1);
close(fds);
printf("normal output");
return 0;

}
```

SLIP NO-15 A:Write aC progra to identify the type of given file using stat() system call

```
#include"stdio.h"
#include"stdlib.h"
#include"unistd.h"
#include"fcntl.h"
#include"sys/stat.h"
#include"sys/types.h"
#include"dirent.h"
int main(int argc,char* argv[])
{
int i;
struct stat S;
if(argc<2)
{
printf("Ente file name on command line");
exit(0);
}
stat(argv[1],&S);
if(S_ISREG(S.st_mode))
printf("File is regular file");
if(S_ISDIR(S.st_mode))
printf("File is Directary file");
if(S_ISSOCK(S.st_mode))
printf("File is soket file");
if(S_ISLNK(S.st_mode))
printf("File is symbolic link file");
if(S_ISBLK(S.st_mode))
printf("File is block file");
if(S_ISCHR(S.st_mode))
printf("File is character file");
if(S_ISFIFO(S.st_mode))
printf("File is pipe file");
return 0;
}
```

```
[fy@localhost Rohit]$ gedit acc15.c
[fy@localhost Rohit]$ gcc acc15.c
/tmp/cc1yn6pr.o: In function `main':
acc15.c:(.text+0x119): undefined reference to `S_ISFZFO'
collect2: error: ld returned 1 exit status
[fy@localhost Rohit]$ gedit acc15.c
[fy@localhost Rohit]$ gcc acc15.c
[fy@localhost Rohit]$ ls -l
total 84
-rw-rw-r--. 1 fy fy  837 Mar 29 15:28 acc13.c
-rw-rw-r--. 1 fy fy 1967 Mar 29 15:27 acc14.c
-rw-rw-r--. 1 fy fy  689 Mar 29 15:48 acc15.c
-rwxrwxr-x. 1 fy fy 8520 Mar 29 15:48 a.out
-rw-rw-r--. 1 fy fy  333 Mar 29 14:32 ass13.c
-rw-rw-r--. 1 fy fy    3 Mar 22 16:16 a.txt
-rw-rw-r--. 1 fy fy   28 Mar 22 15:53 bb.txt
-rwxrwxr-x. 1 fy fy  126 Mar 22 14:49 bhole
-rw-rw-r--. 1 fy fy    8 Mar 22 16:16 b.txt
-rw-rw-r--. 1 fy fy   23 Mar 22 16:16 c.txt
-rw-rw-r--. 1 fy fy    2 Mar 22 16:16 d.txt
-rw-rw-r--. 1 fy fy   14 Mar 22 16:13 p4ab.txt
-rw-rw-r--. 1 fy fy   14 Mar 22 16:13 p4a.txt
-rw-rw-r--. 1 fy fy   35 Mar 22 16:13 p4.txt
-rw-rw-r--. 1 fy fy 2028 Mar 22 16:20 prac4.c
-rw-rw-r--. 1 fy fy  837 Mar 22 15:23 practical1.c
-rw-rw-r--. 1 fy fy  660 Mar 22 15:55 practical2.c
-rw-rw-r--. 1 fy fy 1055 Mar 22 15:59 practical3.c
-rw-rw-r--. 1 fy fy   23 Mar 22 15:14 practicl2.txt
[fy@localhost Rohit]$ ./a.out acc13.c
File is regular file[fy@localhost Rohit]$ mkdir bb
```

SLIP NO-17 B:Write a C prgram to display all the files from current directory which are created in a particular month

```
#include "fcntl.h"
#include "unistd.h"
#include "time.h"
#include "string.h"
#include "dirent.h"
#include "sys/stat.h"
#include "sys/types.h"
int main()
{
char fn[100];
DIR *d;
int num=0;
struct dirent *de;
char *s;
struct stat s1;
printf("Enter month name=");
scanf("%s",fn);
d=opendir(".");
if(d==NULL)
{
printf("Unable to open directory");
exit(0);
}
while((de=readdir(d))!=NULL)
{
stat(de->d_name,&s1);
s=ctime(&s1.st_ctime);
if(strcasestr(s,fn)!=NULL)
{
printf("%s\n",de->d_name);
num++;
}
}
printf("Number file this month %d",num);
```

```
return 0;
}
```

Output:

```
[fy@localhost Rohit AOS]$ gedit sleep5.c
[fy@localhost Rohit AOS]$ gcc sleep5.c
[fy@localhost Rohit AOS]$ ls -l
total 88
-rwxrwxr-x. 1 fy fy 8832 Apr  5 15:10 a.out
-rw-rw-r--. 1 fy fy    3 Mar 22 16:16 a.txt
-rw-rw-r--. 1 fy fy   28 Mar 22 15:53 bb.txt
-rwxrwxr-x. 1 fy fy  126 Mar 22 14:49 bhole
-rw-rw-r--. 1 fy fy    8 Mar 22 16:16 b.txt
-rw-rw-r--. 1 fy fy   23 Mar 22 16:16 c.txt
-rw-rw-r--. 1 fy fy    2 Mar 22 16:16 d.txt
-rw-rw-r--. 1 fy fy   14 Mar 22 16:13 p4ab.txt
-rw-rw-r--. 1 fy fy   14 Mar 22 16:13 p4a.txt
-rw-rw-r--. 1 fy fy   35 Mar 22 16:13 p4.txt
-rw-rw-r--. 1 fy fy 2028 Mar 22 16:20 prac4.c
-rw-rw-r--. 1 fy fy  837 Mar 22 15:23 practical1.c
-rw-rw-r--. 1 fy fy  660 Mar 22 15:55 practical2.c
-rw-rw-r--. 1 fy fy 1055 Mar 22 15:59 practical3.c
-rw-rw-r--. 1 fy fy   23 Mar 22 15:14 practicl2.txt
-rw-rw-r--. 1 fy fy 1287 Apr  5 13:59 sleep1.c
-rw-rw-r--. 1 fy fy 2264 Apr  5 14:17 sleep2.c
-rw-rw-r--. 1 fy fy 1244 Apr  5 14:51 sleep3.c
-rw-rw-r--. 1 fy fy  919 Apr  5 15:00 sleep4.c
-rw-rw-r--. 1 fy fy  539 Apr  5 15:10 sleep5.c
[fy@localhost Rohit AOS]$ ./a.out apr
Enter month name=apr
sleep1.c
sleep2.c
sleep3.c
sleep4.c
sleep5.c
a.out   Number file this month 8
```